June 2026 Plan — RTL build + bit-exact co-simulation

Aligned with the research plan (§9.3): *implement* `item_memory`, `bundle_unit`, `pruning_mask`, `popcount_am` *RTL + co-sim passing; D parameterisation verified.*

Goal (one sentence)

> Implement the four new RTL modules and have them pass **bit-exact co-simulation** against `hdc_ref.py`, with **D** and **bundle counter width (CNT_W)** parameterisation verified — no board, no power, no experiments yet.

Starting point (already done in May)

- `xor_permute_top.sv` + `permute_stage.sv` + self-checking testbench.
- `hdc_ref.py` golden reference (smoke tests pass).
- 1000 bind+permute co-sim vectors in `python_ref/vectors/`.
- Item-memory and example pruning-mask `.mem` files in `python_ref/mem_files/`.
- Frozen software baseline: 90.30 % ± 0.13 spatial at D=1024 (protocol P-may2026).

Week-by-week

Week 1 — Co-sim harness + `item_memory`

#	Task	Output
J1	Automated co-sim harness: TB reads `vectors/*.hex` + `mem_files/`, runs RTL, compares to golden, prints PASS/FAIL; wrap as one `.do` script	`tb/cosim_*.sv`, `sim/run_cosim.do`
J2	`item_memory.sv` — channel iM + continuous level CiM via `$readmemh`, parameterised `D` (replaces `simple_bind_rom` placeholder)	`rtl/item_memory.sv`
J3	TB: `item_memory` lookups vs `hdc_ref` exported memories, bit-exact	`tb/tb_item_memory.sv`

Week 2 — `bundle_unit` + `popcount_am`

#	Task	Output
J4	`bundle_unit.sv` — thresholded majority, tie→0, parametric `CNT_W`	`rtl/bundle_unit.sv`

#	Task	Output
J5	`popcount_am.sv` — masked Hamming distance to K class prototypes + argmin; parametric `D`, `K`	`rtl/popcount_am.sv`
J6	Extend `generate_vectors.py` to emit bundle + AM cases; TBs vs golden	new vectors + 2 TBs

Week 3 — `pruning_mask` + integration

#	Task	Output
J7	`pruning_mask.sv` — per-bit AND before popcount (mask from `mem_files/`)	`rtl/pruning_mask.sv`
J8	Integrate bind → permute → bundle → (mask) → popcount into a classify-pipeline top	`rtl/hdc_classify_top.sv`
J9	End-to-end co-sim: encode EMG windows in Python, compare RTL **decisions** bit-exactly at D=1024	end-to-end TB + report

Week 4 — D parameterisation + regression

#	Task	Output
J10	Verify `D ∈ {256,512,1024,2048}` and `CNT_W` variants match golden	sweep log
J11	One-command regression (compile + all TBs + end-to-end → PASS/FAIL)	`sim/run_regression.do`
J12	Tag June milestone; write June results PDF note	`python_ref/notes/june_cosim.pdf`

Explicitly NOT in June

- **July:** `hdc_stream_wrapper.sv` (AXI-Stream + DMA) and board bring-up.
- **August:** Hook A Pareto + Twist 1 (informed vs random) + Twist 2 (cross-subject) + power.
- **September:** paper draft → DATE submission.

"June done" checklist

- [ ] `item_memory`, `bundle_unit`, `pruning_mask`, `popcount_am` implemented
- [ ] each passes bit-exact co-sim vs `hdc_ref.py`
- [ ] full pipeline matches Python decisions at D=1024
- [ ] D parameterisation verified across ≥3 values
- [ ] one-command regression passes